

DAY 03 LECTURE · 90 MINUTES

Living in the Cloud.

SUBTITLE

Renting computing by the
slice

TODAY

Virtualization · The models

THEN

Regions · Scaling · The Big
Three

RECAP

Yesterday: how it travels

TODAY'S EXPERIMENT

You have an idea for an app. Ninety minutes from now it is running **on every continent.** And you never buy a single computer.

YOU ALREADY LIVE IN THE CLOUD – YOUR PHOTOS, YOUR CHATS, THIS LECTURE. TODAY YOU LEARN TO BUILD IN IT: WE FOLLOW YOUR APP FROM FIRST SERVER TO FIRST INVOICE.

THE CLOUD IS SOMEONE ELSE'S COMPUTER

You own **not one** of these machines. You rent a sliver of them, by the second.

Five steps, **one launch.**

YOUR APP'S JOURNEY, TODAY'S LECTURE

§ 01

How do you rent half a computer?

Virtualization, the trick that turns one machine into many.

§ 02

What exactly are you renting?

Infrastructure, platform, or finished software. IaaS, PaaS, SaaS.

§ 03

Where does your slice live?

Regions and availability zones, and why geography still matters.

§ 04

What happens when everyone shows up?

Elasticity, auto-scaling, and computing that bills by the second.

§ 05

Who are you renting from?

The Big Three, lock-in, and the bill at the end of the month.

— SECTION ONE

01

The great **trick.**

None of the cloud is possible without one clever sleight of hand: **virtualization.**

• YOUR APP • STEP 1 • IT NEEDS A COMPUTER YOU WILL NOT BUY

THE IDEA IN ONE LINE

Take one physical computer and make it **pretend to be many**. Each pretend computer thinks it is alone.

ONCE YOU CAN SPLIT A MACHINE CLEANLY, YOU CAN RENT OUT THE
PIECES. THAT IS THE WHOLE BUSINESS OF THE CLOUD.

A COMPUTER INSIDE A COMPUTER

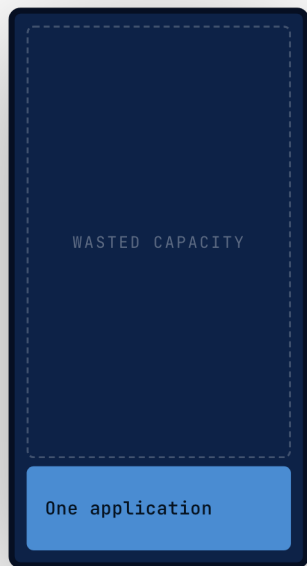
The **virtual machine**.

A virtual machine is software pretending to be hardware. It has its own operating system, its own memory, its own storage. As far as it knows, **it is a real computer**.

Think of one big building divided into **apartments**. Each tenant has their own front door and never sees the others, even though they share the same walls and foundation.



FIG. APARTMENTS IN ONE BUILDING



UTILISATION OF ONE SERVER

16%

A single app barely troubles a whole server, yet you pay for the entire box. Across a fleet, most of the metal sits **idle and expensive**.

Bare metal

Virtualised

THE THING THAT KEEPS THEM APART

Every building needs a landlord.

The **hypervisor** is a thin layer of software that sits on the real machine and hands out slices of it: a little processor here, some memory there.

Its most important job is keeping tenants **strictly separated**, so one noisy neighbour can never see, slow, or crash the others.

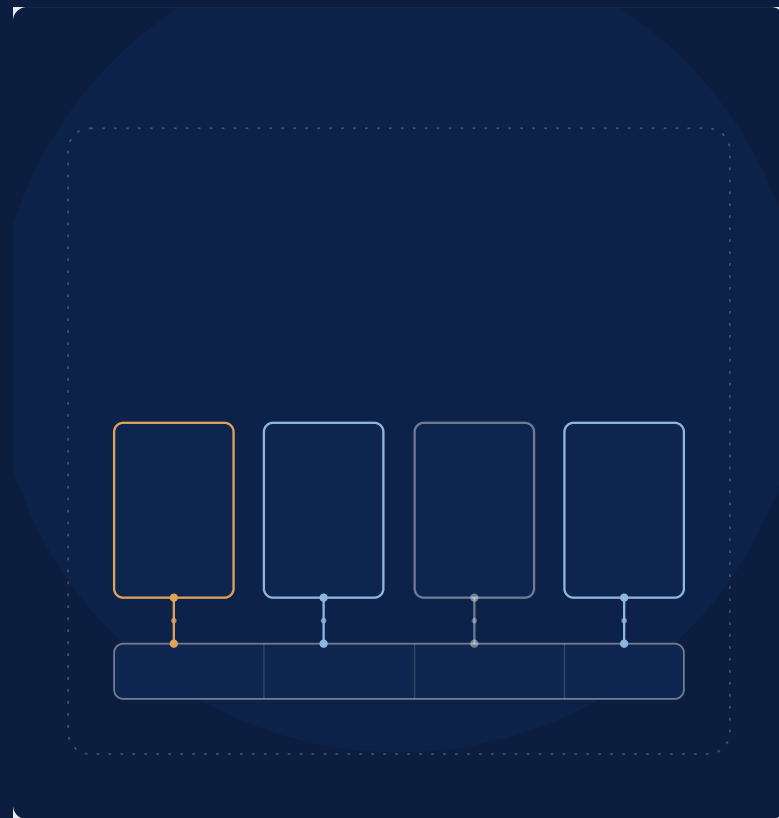


FIG. THE HYPERVISOR LAYER

A LIGHTER WAY TO PACK

Then came **containers**.

A virtual machine ships a whole operating system, which is heavy. A **container** ships only your app and the few things it needs, sharing the OS underneath.

Lighter, faster to start, easy to copy. The same app box runs identically on your laptop and on a server **on the other side of the world**.

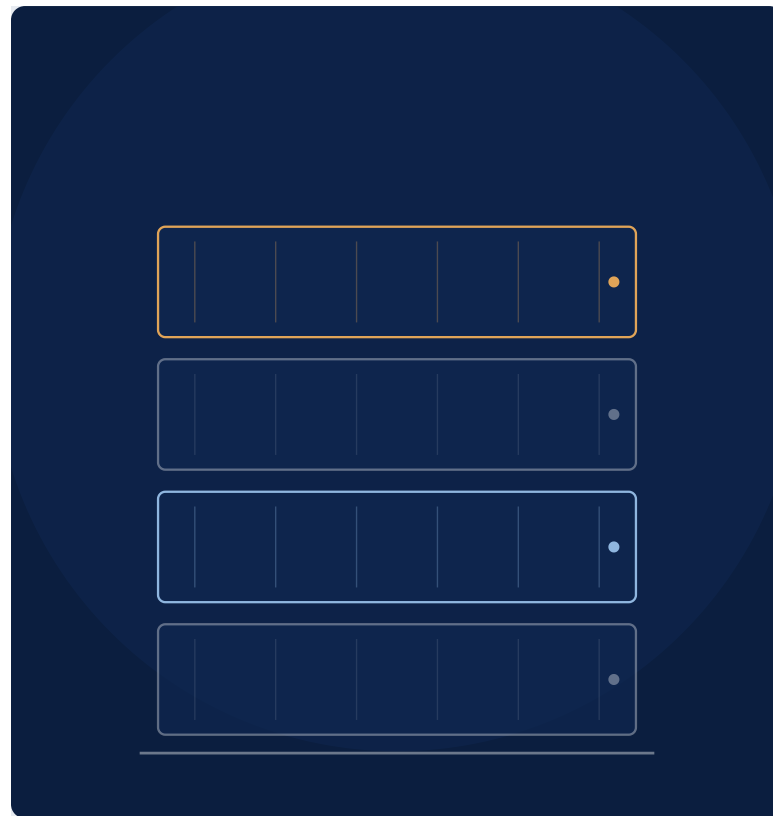


FIG. SHIP THE APP, NOT THE BUILDING

Two ways to **slice a machine.**

BOTH USEFUL, FOR DIFFERENT JOBS

VIRTUAL MACHINE

A whole **house.**

Its own operating system, fully isolated, heavier to move. Best when you need a complete, separate computer.

CONTAINER

A packed **box.**

Just the app and its essentials, sharing the OS. Starts in a blink, and you can run hundreds on one host.

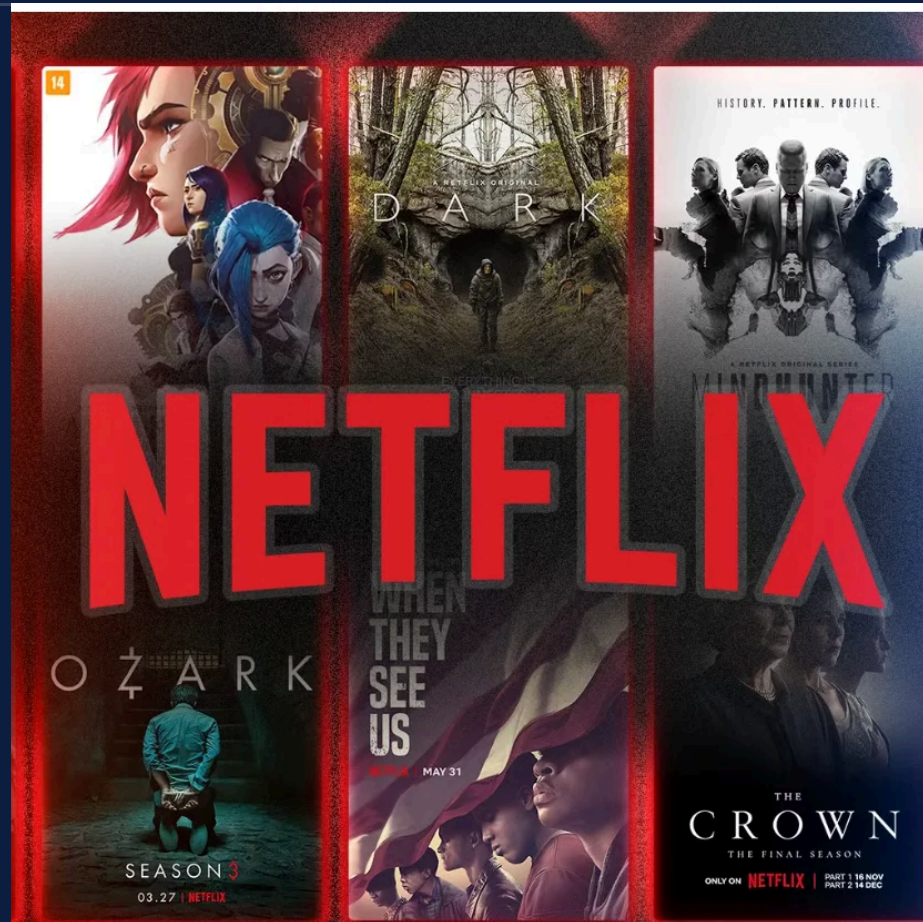
• STORY • 2008 TO 2016

Netflix tore down its own data centres **on purpose.**

In 2008, a failure in Netflix's own data centre stopped DVDs shipping for three days. They drew a hard conclusion: **running their own machines was the risk.**

Over the next years they moved everything onto rented cloud servers. By 2016 they ran **no data centres at all**, yet streamed to the whole planet.

The lesson: for many, the safest, most flexible machine room is the one you do not own.



— SECTION TWO

02

Renting **computing.**

You can rent the bare machine, the whole platform, or the finished software.
How much do you want to manage?

• YOUR APP • STEP 2 • DECIDE HOW MUCH YOU MANAGE

THE SHIFT THAT STARTED IT ALL

You do not run a generator in your basement. You plug into the wall. **Cloud is computing sold the same way:** as a utility, by the unit.

STOP BUYING MACHINES YOU HAVE TO OWN, POWER AND REPLACE.
START PAYING ONLY FOR THE COMPUTING YOU ACTUALLY USE.

Your app needs a server. **Two eras, one race.**

START BOTH CLOCKS

WHAT “GETTING A COMPUTER” USED TO MEAN, AND MEANS NOW

2005 • BUY THE SERVER

Budget approved

Servers ordered

Delivery & customs

Racked, wired, cooled

OS installed • live

0 days

WAITING

TODAY • RENT THE SLICE

Sign in to the console

Pick a size & region

Click “Launch” • live

0s

WAITING

HOW MUCH DO YOU MANAGE?

00 On-premises

IaaS Infrastructure

PaaS Platform

SaaS Software

■ YOU MANAGE ■ PROVIDER MANAGES

Applications

YOU

Data

YOU

Runtime

YOU

Middleware

YOU

Operating system

YOU

Virtualisation

PROVIDER

Servers

PROVIDER

Storage & networking

PROVIDER

Rent raw virtual machines. The provider keeps the metal alive; **you still run the OS and everything above it.**

The same three, **in plain words.**

MORE CONVENIENCE AS YOU GO DOWN

IaaS

Rent the raw machines

Virtual servers you set up yourself. Maximum control, maximum work. Like leasing an empty workshop.

PaaS

Rent the workbench

Bring your code; the provider runs the servers and tools underneath. You build, they maintain.

SaaS

Rent the finished product

Just log in and use it. Email, docs, video calls. Someone else worries about everything else.

THE ANALOGY THAT FINALLY MAKES IT STICK

Want pizza? Make it from scratch. Buy a frozen one. Get it delivered. Or **sit down at a restaurant.**

SAME DINNER, FOUR LEVELS OF HELP. ON-PREMISES, IAAS, PAAS, SAAS. THE MORE YOU HAND OVER, THE LESS YOU HAVE TO DO, AND THE LESS YOU CONTROL.

• STORY • DROPBOX

Sometimes the smart move is **leaving.**

Dropbox grew up on rented cloud storage. But at their enormous scale, the rent became the single biggest cost. So they did the unfashionable thing: they **built their own data centres** and moved most data back.

The shift reportedly saved them **tens of millions of dollars** a year.

The lesson: the cloud is brilliant, not magic. At a certain size, owning can beat renting. Know which you are.



Dropbox

— SECTION THREE

03

Where your **slice** lives.

“The cloud” sounds placeless. It is the opposite. Every byte sits somewhere specific.

• YOUR APP • STEP 3 • CHOOSE WHERE ON EARTH IT LIVES

A WORD THAT HIDES THE TRUTH

When you launch a server, the very first thing you choose is **where on Earth** it runs. The cloud has a map, and you pick a pin.

THAT CHOICE QUIETLY DECIDES THREE THINGS: HOW FAST IT FEELS,
WHICH LAWS APPLY TO IT, AND WHAT IT COSTS.

THE BIG PINS ON THE MAP

A **region** is a place.

A region is a cluster of data centres in one part of the world: Bahrain, Frankfurt, Singapore, Northern Virginia. Each provider has **dozens**.

Put your service near your users and it feels **instant**.
Put it on the far side of the planet and yesterday's lesson kicks in: the speed of light makes you wait.

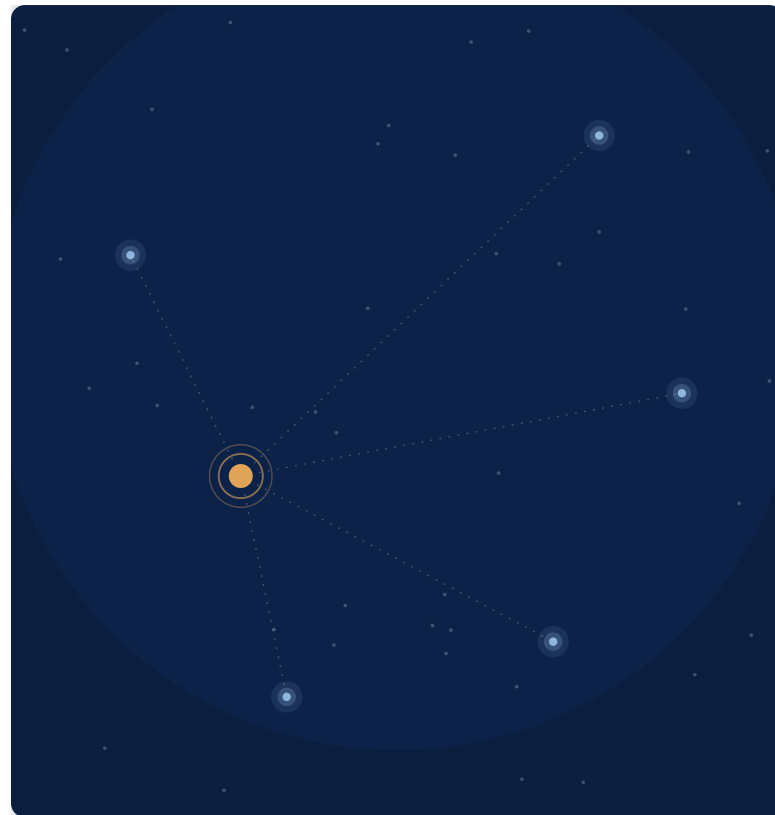


FIG. PICK A PIN

INSIDE ONE REGION

Several buildings, **one address.**

A region is not a single building. It is split into **availability zones**: separate data centres, kilometres apart, with their own power and cooling.

Spread your service across zones and a fire, flood or outage in one cannot take you down. **The others simply carry on.** This is yesterday's redundancy, drawn on a map.

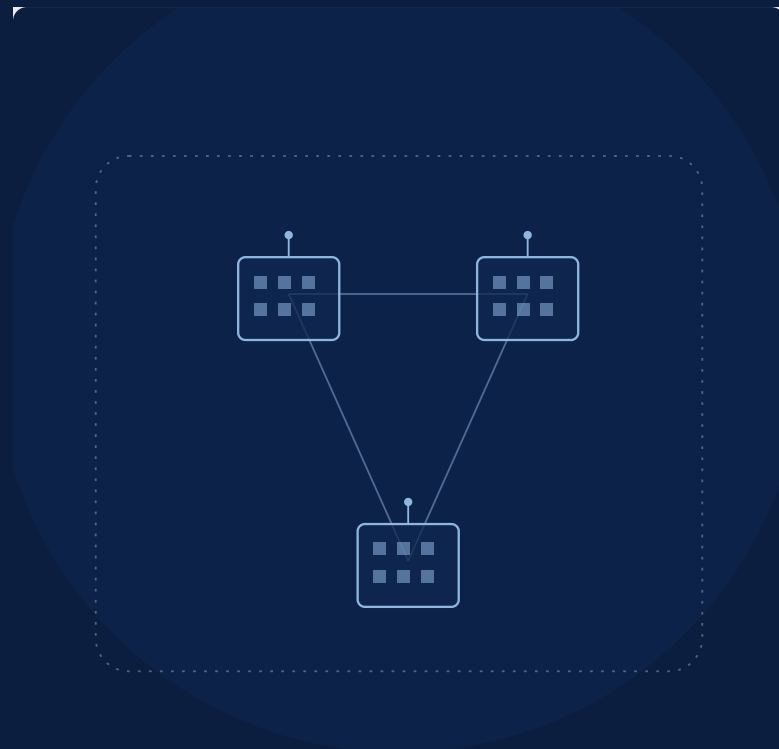


FIG. ZONES WITHIN A REGION

FROM JEDDAH, PICK A REGION

Bahrain

me-south-1

Frankfurt

eu-central-1

Ireland

eu-west-1

Singapore

ap-southeast-1

N. Virginia

us-east-1

11_{ms}

ROUND TRIP FROM JEDDAH

3

AVAILABILITY ZONES

Zone A

Data centre A

HEALTHY

Zone B

Data centre B

HEALTHY

Zone C

Data centre C

HEALTHY

Still online. Traffic shifts to the healthy zones.

CLICK A ZONE TO KNOCK IT OFFLINE

Your data has **a passport.**

WHY REGION CHOICE IS ALSO A LEGAL CHOICE

01

The law of the land applies

Data stored in a country usually falls under that country's rules on privacy, access and disclosure.

02

Some data is not allowed to leave

Health records, banking and government data often must, by law, stay inside national borders.

03

So providers build local regions

A region in Saudi Arabia, in Germany, in India. Sovereignty, increasingly, is built into the map.

• STORY • US-EAST-1

When one region sneezes, the web catches cold.

There is one region in Northern Virginia that is the oldest, cheapest and most popular. So much of the internet quietly defaults to it that when it stumbles, **the outages make headlines worldwide.**

Shops, games, smart doorbells, even other companies' control panels go dark together, all because they sat in **the same place.**

The lesson: the cloud gives you the whole planet to spread across. Convenience tempts everyone to crowd into one corner of it.



— SECTION FOUR

04

Elastic & serverless.

The cloud's real superpower is not renting machines. It is renting **exactly as many as you need, exactly when you need them.**

• YOUR APP • LAUNCH DAY • THE CROWD IS COMING

THE SHAPE-SHIFTING MACHINE ROOM

A shop hires extra staff only for the lunch rush. The cloud does the same with servers: **grow for the crowd, shrink when they leave.**

NO MORE BUYING FOR YOUR BUSIEST DAY AND PAYING FOR IT EVERY QUIET NIGHT. THE INFRASTRUCTURE BREATHES WITH DEMAND.

One day of traffic, **matched** in real time.

SERVICES FOLLOW THE CROWD, HOUR BY HOUR

1

SERVICES RUNNING NOW

- Traffic
- Servers running

▶ [Replay the day](#)



THE METER IS ALWAYS RUNNING

by the second

You are no longer billed for a machine you bought. You are billed for the computing you **actually used**, often measured by the second. Use nothing overnight, pay close to nothing.

THE VANISHING SERVER

Stop thinking about **servers** at **all**.

With **serverless**, you hand over a single function of code and a trigger. The provider runs it the instant it is needed, on whatever machine is free, then forgets it.

There are still servers, of course. You just never see, choose or manage them. **You pay only for the milliseconds your code runs.** Today's lab ends exactly here.

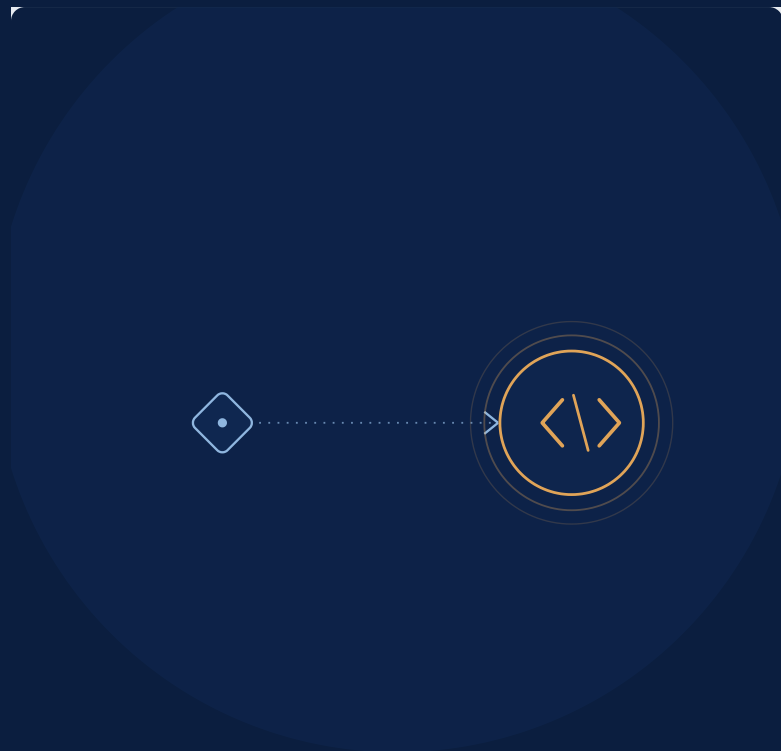


FIG. JUST THE FUNCTION

NOTHING IS FREE, INCLUDING CONVENIENCE

A function that has sat idle has to wake up first, a tiny pause called a **cold start**.

AND HANDING EVERYTHING TO ONE PROVIDER IS EASY TO START AND HARD TO UNDO. EVERY LAYER OF CONVENIENCE TRADES AWAY A LITTLE CONTROL. THAT IS THE DEAL, AND IT IS WORTH UNDERSTANDING BEFORE YOU SIGN IT.

Feel a **cold start**.

CALL YOUR SLEEPING FUNCTION, THEN CALL IT AGAIN

CALL THE FUNCTION

Let it sit idle a while

— ms
LAST RESPONSE TIME

■ WAKING YOUR CODE ■ RUNNING YOUR CODE

• STORY • 2016

A game outgrew its **wildest forecast** in days.

When Pokémon Go launched, players appeared **far faster than anyone planned for**. Real-world demand blew past the team's most optimistic estimate many times over.

Because it ran on elastic cloud infrastructure, the servers simply **multiplied to meet the crowd**, then eased back down. A decade earlier, this would have meant weeks of buying and wiring machines that arrived too late.

The lesson: elasticity turns a viral moment from a disaster into a good day.



— SECTION FIVE

05

The **Big Three.**

Almost everything you have learned today is rented from one of **three** companies.

• YOUR APP • MONTH ONE • THE INVOICE ARRIVES

Three landlords for **most** of the internet.

YOU MET THEM ON DAY 1. HERE IS HOW THEY SELL

AWS

Amazon Web Services

The first mover and still the broadest. If a cloud service can exist, AWS probably already rents it.

Azure

Microsoft Azure

The favourite of large enterprises, woven into the Microsoft tools companies already run.

GCP

Google Cloud

Strongest in data and AI, built on the same foundations that run Search and YouTube.

THE GOLDEN HANDCUFFS

Easy to move **in**.

Each provider offers wonderful tools that work beautifully **together**, and only together. The more of them you use, the more your whole system is shaped around **that one provider**.

Leaving later means rebuilding a great deal from scratch. The cloud welcomes you warmly, and the welcome is also **a quiet form of gravity**.

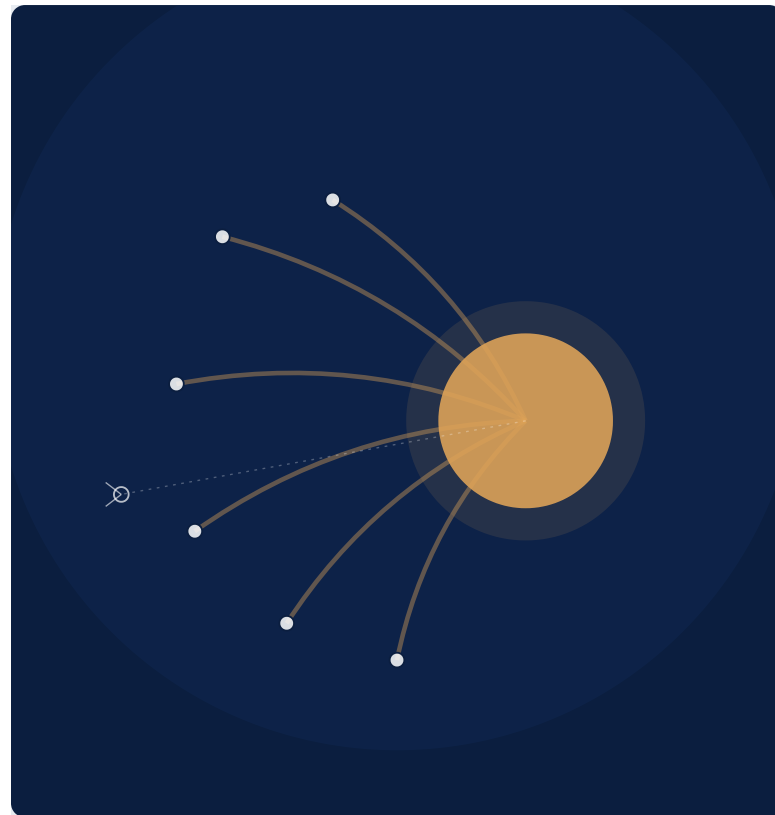


FIG. IN IS EASIER THAN OUT

So why not use **all three?**

MULTI-CLOUD, AND ITS HONEST TRADE-OFF

THE PROMISE

No single **landlord.**

Spread across providers and you dodge lock-in, survive one provider's outage, and shop each for its best service.

THE PRICE

Twice the **complexity.**

Every provider does things differently. Running well across several means more skills, more tools, and more ways to get it wrong.

Leave something running. See what it costs.

EIGHT RENTED GPUS, ONE FORGOTTEN WEEKEND

CLOSE YOUR LAPTOP · FRI
18:02

FRI 18:02

\$0

METERED THIS WEEKEND



● BUDGET ALARM · NOT SET

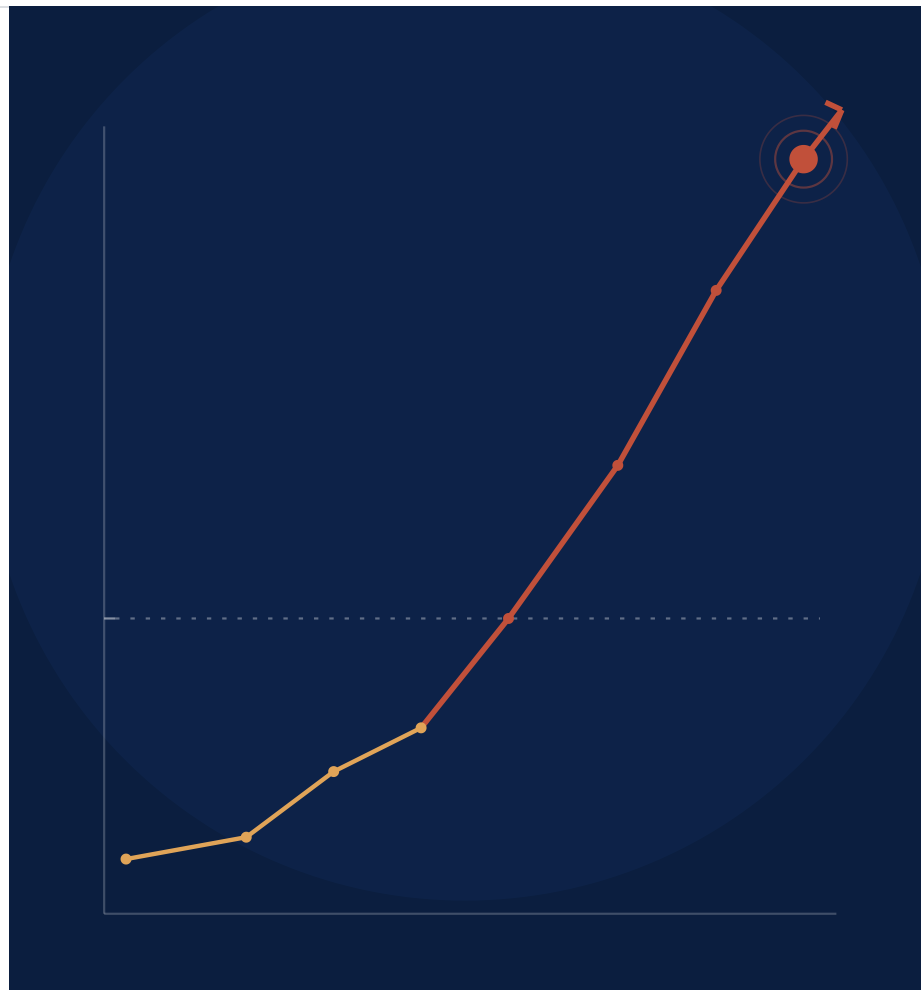
• STORY • THE METER NEVER SLEEPS

A single line of config, a terrifying invoice.

It is a rite of passage: a developer leaves a powerful machine running over a weekend, or a runaway loop calls a paid service millions of times. On Monday, the bill is **thousands of dollars**.

The same meter that charges you almost nothing when idle charges **relentlessly** when something goes wrong, with no one watching.

The lesson: renting by the second is wonderful and unforgiving. In the cloud, mistakes have a price, in real money.



— THE REFRAME

You stopped buying **computers** and started renting **computing**, by the slice, by the second, anywhere on Earth.

Your app is live on every continent, and you never touched a single machine.
But everything it does still happens in a handful of giant buildings, far away.

Tomorrow: what happens when we push the computing back out, all the way to your hand. Out to the edge.